

Diagnosis

- Introduction
- Logic Diagnosis
- Scan Chain Diagnosis
- Failure Analysis *(not in exam)
- Conclusions



Physical Failure Analysis, PFA

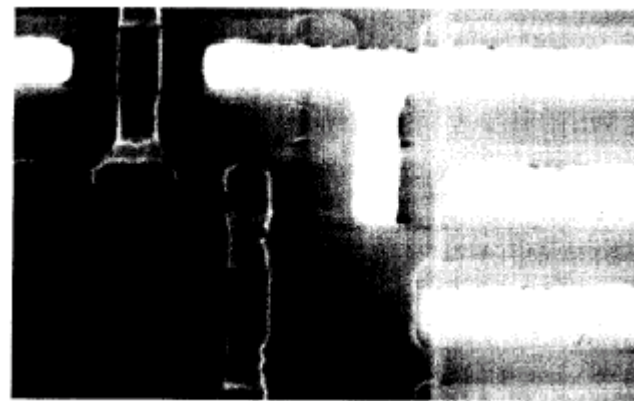
- **What is PFA**
 - ◆ Open chip package, locate and examine failure mode of defect
 - ◆ Aka.
 - * *Failure Analysis (FA)*
- **Why PFA**
 - ◆ Verify diagnosis results
 - ◆ Improve fab/test/design
 - * Improve yield
 - ◆ Improve design or test techniques

Failure Analysis

- Examine failure mode of defects
 - ◆ Failure mechanism can be inferred
- Usually requires lots of experience and equipments
- Example FA pictures [Soden 89]
 - ◆ Scanning Electron Microscope picture (Voltage Contrast)
 - ◆ Failure mechanism: photoresist failed to adhere
 - ◆ Failure mode: open circuit



good



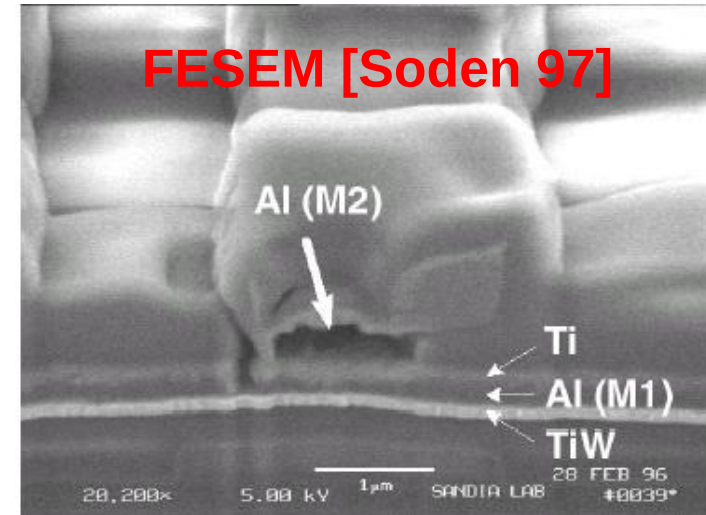
defective

FA Techniques

- Imaging without power
- Imaging with power
- Physical process
 - ◆ Etching, cross sectioning
- See more details [Soden D&T 97]

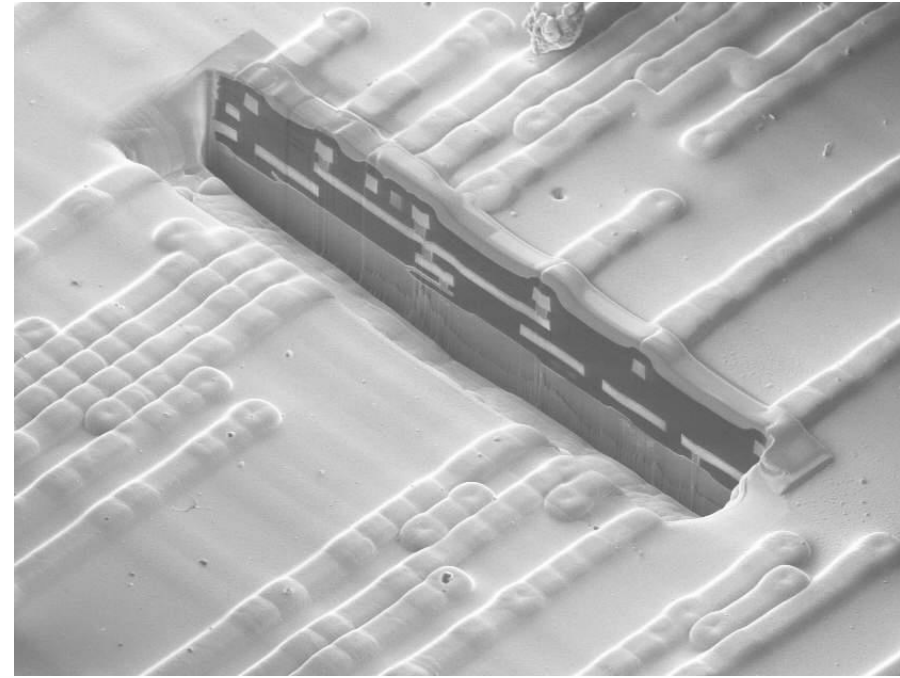
Imaging Techniques (no power)

- Optical microscope (Resolution limit :0.2 to 0.3 μm)
 - ◆ One useful technique: Scanning Optical Microscope (SOM)
 - * focus infrared laser spot scanning over sample
 - + Silicon transparent to infrared
 - Resolution limit: 1 μm
- Scanning Electron Microscope (SEM)
 - ◆ Field emission Scanning Electron Microscope (FESEM)
 - * Resolution limit: 2 to 5 nm
 - ◆ Transmission Electron Microscope (TEM)
 - * Resolution: sub nm
- Scanning probe microscope (SPM)
 - ◆ Sharp probe tip scans sample surface
- Focus Ion Beam (FIB)



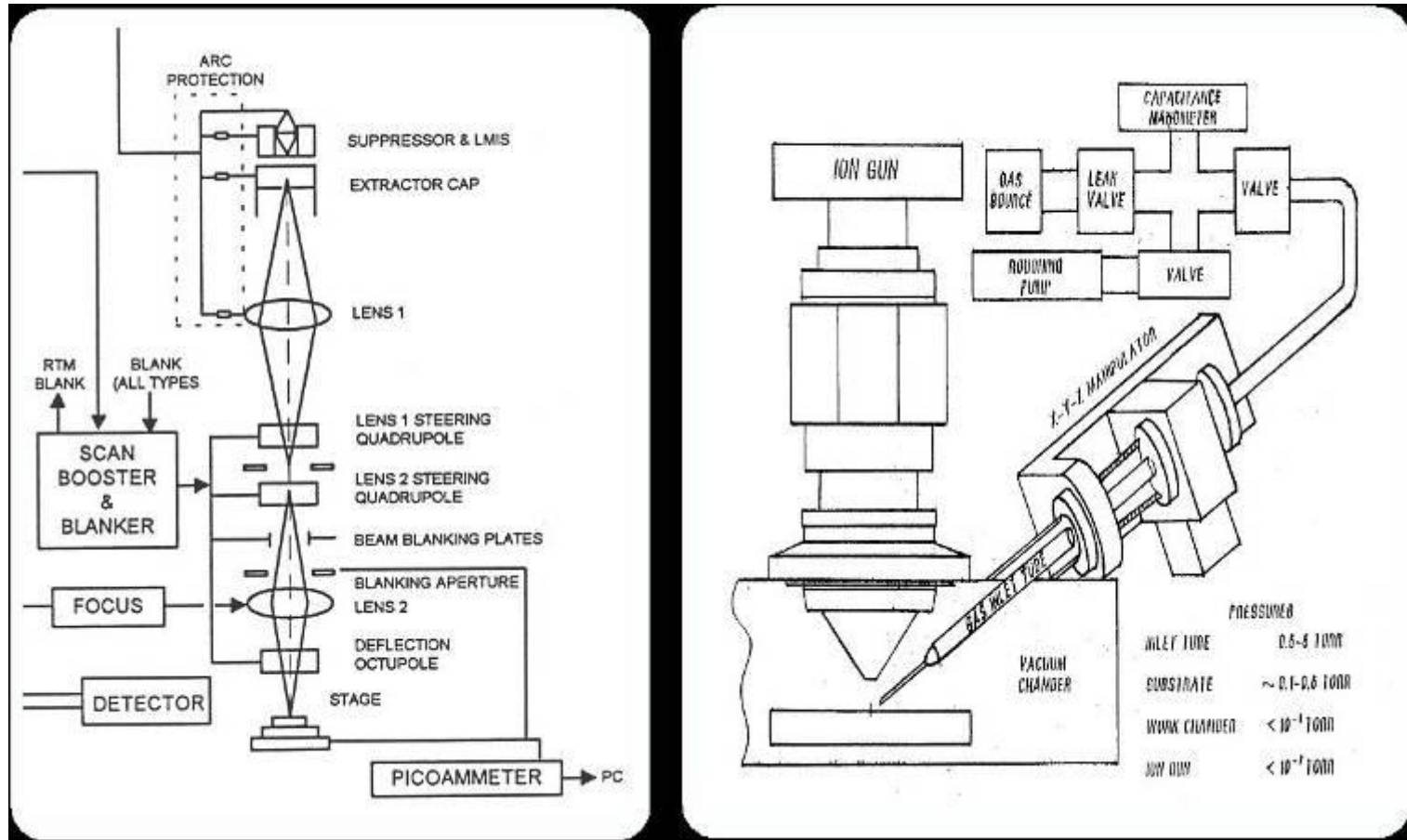
Focus Ion Beam (FIB)

- Focused beam of Ga^+ ion
- Multiple usages
 - ◆ Image
 - ◆ Circuit editing (down to 45 nm),
 - ◆ FA cross-sectioning (wider than 200 μm) with e-beam imaging and ion channeling contrast
 - ◆ Ion Channeling Contrast for grain imaging of crystalline materials
 - ◆ Passive Voltage Contrast for FA
 - ◆ Micro-machining for MEMS
 - ◆ TEM sample preparation, both cross-sectional and plan-view (min. size 18 nm)
 - ◆ Ion implantation (Ga^+)



[image courtesy of SK Lu from FJU]

FIB system

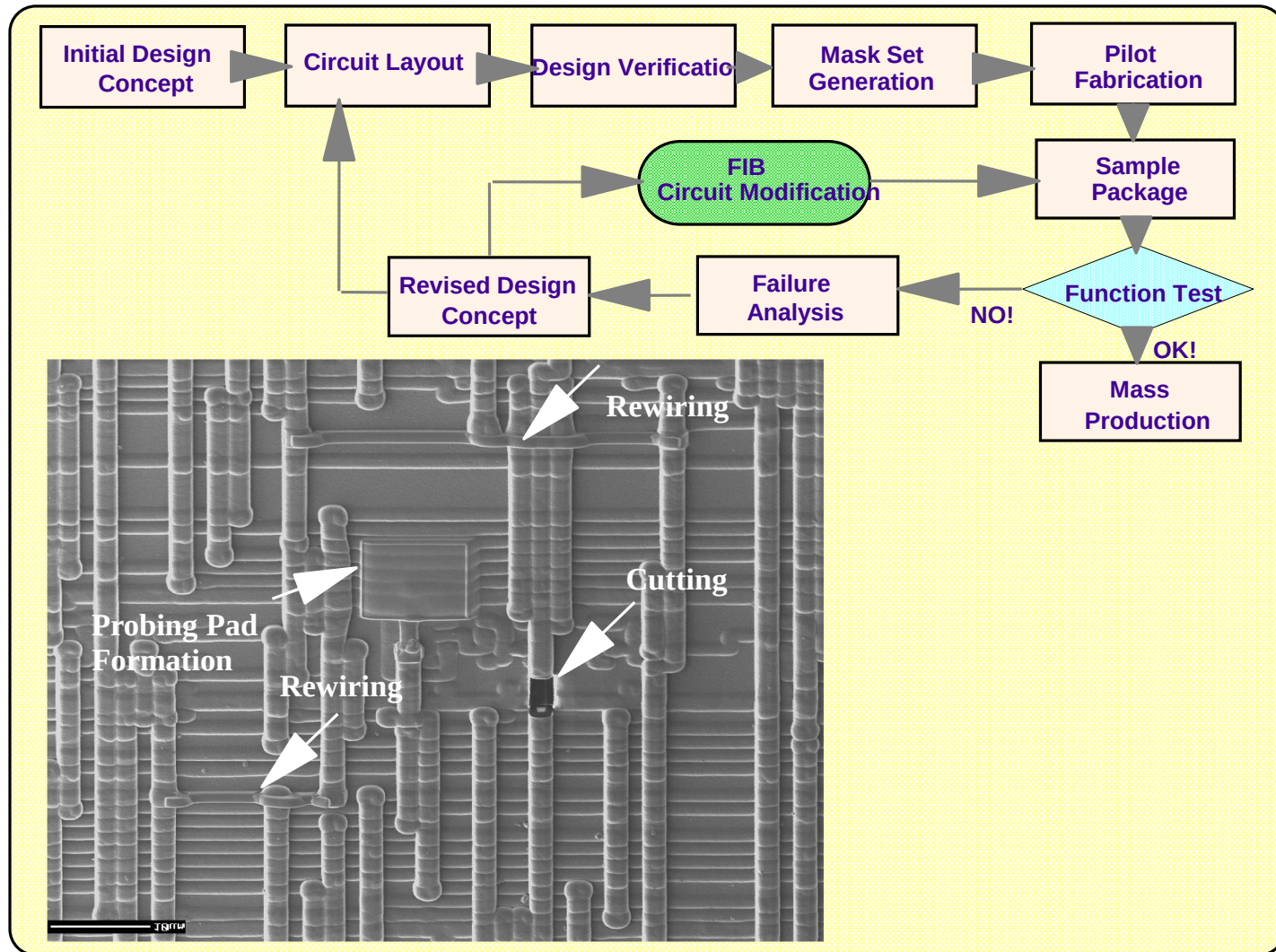


FIB photo

- Source : MA TEK



FIB Circuit Editing



Imaging Techniques (powered)

- Light emission (aka photo emission microscope)
 - ◆ Apply test patterns, capture light by camera
 - ◆ Very commonly used
 - + Feature size independent
 - + Infrared light enable analysis form die backside

Light emission microscopy [Soden 97]

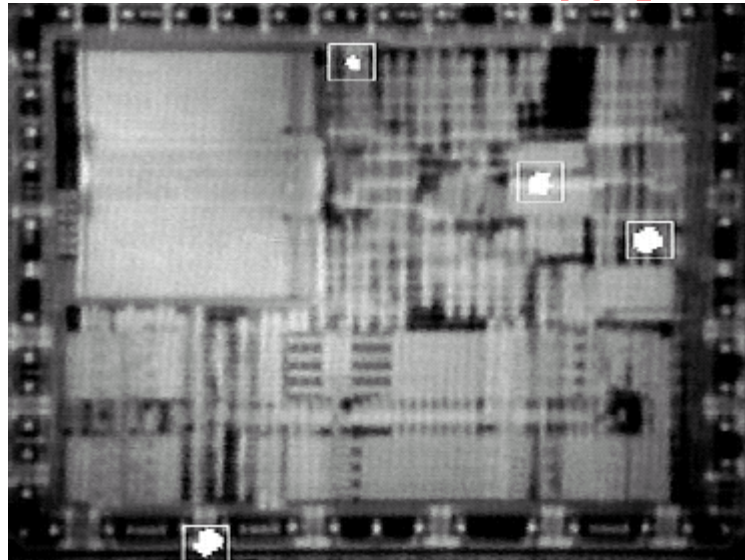


Figure 2. Low-magnification light emission microscopy image of a 32-bit microprocessor with four gate oxide shorts (bright areas in the four boxes).

Imaging Techniques (powered)

- Voltage Contrast (VC) and Electron Beam (E beam)
 - ◆ Probe change of signals as well as voltage measurement
 - + Nondestructive
 - Cannot probe Flip chip package
 - More noise for small geometry and high frequency
 - Difficult to probe lower layer of metals
- Resistive Contrast Imaging (RCI)
 - ◆ SEM that generates relative resistance image between two nodes
 - ◆ Useful for open defects

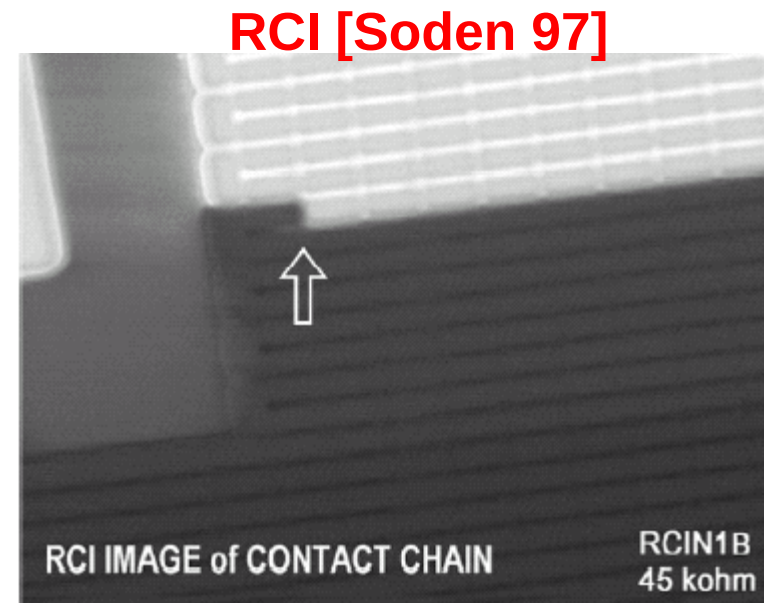


Figure 11. RCI image of contact chain with contrast discontinuity at high-resistance contact (arrow).

Imaging Techniques (powered)

- **Charge and Light-Induced Voltage Alteration (CIVA/ LIVA)**
 - ◆ **Useful for open defects**
- **Scanning Probe Microscope**
- **FIB**
- **Thermal imaging**
 - ◆ **Detects “hot spots” caused by defects**
 - ◆ **Commonly used techniques:**
 - * **Liquid crystal**
 - * **Infrared thermography**

Integrated E-Beam Probe System

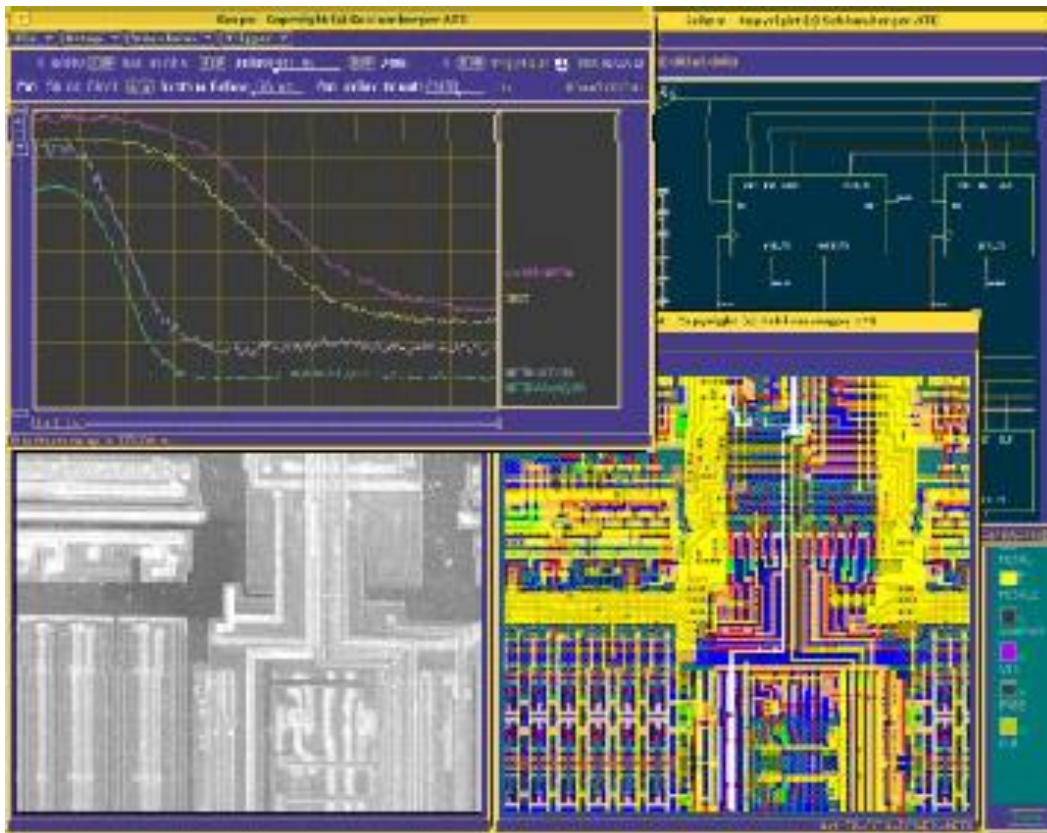
- Show different view all together

Scope

Schematic

SEM

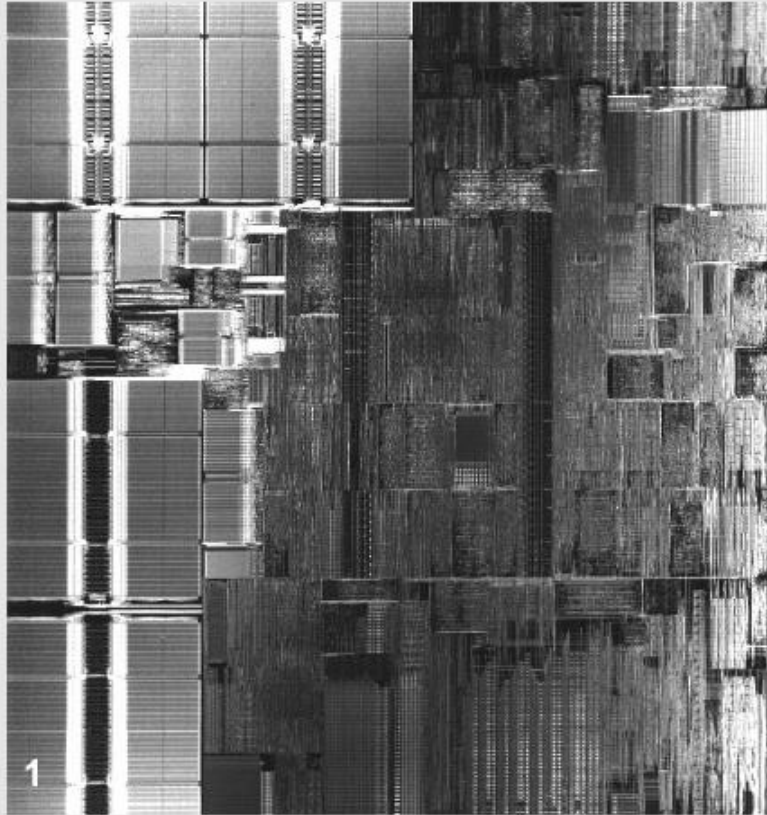
Layout



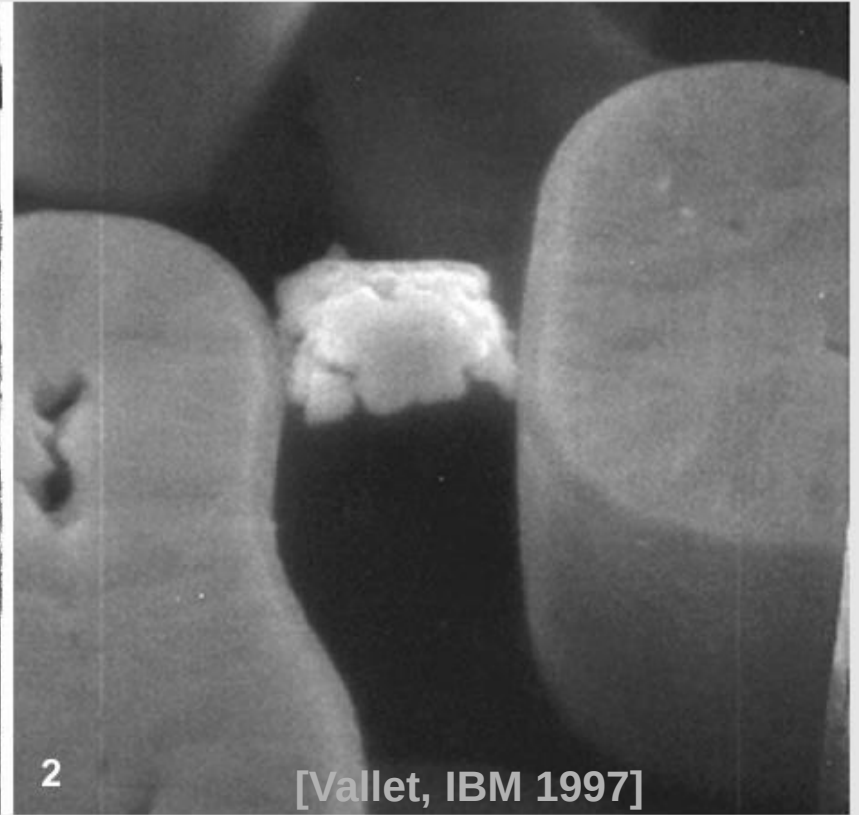
[image courtesy of SK Lu from FJU]

Challenges of Failure Analysis

- # 1 challenge: Looking for a needle in haystack ...
 - ◆ Good software diagnosis is very important



5M Transistor Microprocessor, 196mm²



0.2x0.4μm², nickel particle causing short

- FA equipment are expensive

FA Equipment Costly

- **Cost of useful FA tools (USD)**
 - ◆ Microscope: 50K
 - ◆ X-ray: 150K
 - ◆ CSAM: 150K
 - ◆ Decapsulator: 40K
 - ◆ Automatic curve tracer: 100K
 - ◆ SEM: 200K
 - ◆ LEM: 150K
 - ◆ Probe station: 100K(semi-auto)
 - ◆ FIB: 500K
 - ◆ E-Beam probe: 300K

Diagnosis

- Introduction
- Logic Diagnosis
- Scan Chain Diagnosis
- Failure Analysis
- Conclusions



Conclusion

- Diagnosis important for yield improvement and quick time to market
- Categories
 - ◆ Logic Diagnosis
 - * Model-based diagnosis: SSF, delay fault
 - * Modeless diagnosis: inject and cure, SLAT
 - ◆ Scan chain diagnosis
- Diagnosis was usually embedded in ATPG
 - ◆ Mentor Graphics: Fastscan
 - ◆ Synopsys: Tetramax
 - ◆ Cadence: Encounter Test
- Now, diagnosis also available as independent tool
 - ◆ Mentor Graphics: Yield assist

References

- [Abramovici 1984] M. Abramovici, P. R. Menon, and D. T. Miller, “Critical Path Tracing: An Alternative to Fault Simulation,” IEEE Design and Test of Computers, 1984.
- [Akers 90] S.B. Akers ; S. Park ; B. Krishnamurthy ; A. Swaminathan, “Why is less information from logic simulation more useful in fault simulation?” IEEE Intl test conference , 1990.
- [Bartenstein 2001] T. Bartenstein, L. Huisman and et al, “Diagnosing combinational logic designs using the single location-at-a-time (SLAT) paradigm,” IEEE ITC, pp.287-296, 2001.
- [Girard 92] P. Girard, “A Novel Approach to Delay-fault diagnosis,” IEEE ICCAD, pp.357-360, 1992.
- [Guo 01] R. Guo, S. Venkataraman, “Technique for Fault Diagnosis of Defects in Scan Chains,” IEEE ITC, pp268, 2001.
- [Huang 97] SY Huang, KT Cheng, KC Chen, DI Cheng “ ErrorTracer: A Fault Simulation Based Approach to Design Error Diagnosis,” Proc. of Int'l Test Conf, 1997.
- [Huang 08] Y. Huang, R. Guo, WT Cheng, and JCM Li, “Survey of Scan Chains Diagnosis,” D&T of Computers, May/June pp240-248, 2008.
- [Li 05] JCM Li, “Diagnosis of Single Stuck-at Faults and Multiple Timing Faults in Scan Chains,” IEEE TVLSI, Vol.13. No.6, pp.708-718, 2005
- [Kundu 93] S. Kundu, “On diagnosis of faults in a scan-chain,” IEEE VTS pp.303-308, 1993.
- [Smith 05] A. Smith and et al, “Fault Diagnosis and Logic Debugging Using Boolean Satisfiability,” IEEE TCAD Vol.24, No. 10, pp.1606-1620, 2005.